

duino UNO R3 (MOD2) and buzzer (B1) to sense the liquid level. For the sensor detecting liquid, Arduino gives a signal to the buzzer and the inbuilt LED for detecting the same.

Table 2 shows the connections

between XKC-Y25 NPN sensor and the Arduino board. Fig. 4 shows the wiring of the proposed liquid level detector on breadboard.

The operating voltage (VCC) of sensor XKC-Y25 NPN ranges

between 5V and 24V DC and the current consumption is around 5mA. Its high-level output voltage equals VCC and the low-level output voltage is zero. Its output current ranges between 1 and 50mA, while its response time is around 500ms. Operating temperature of the sensor is from 0 to 105°C. Its range of induction (sensitivity) lies between 0 and 13mm and humidity 5 to 100%. The sensor is made of ABS plastic and has IP67 level of waterproof protection.

Software

The proposed setup involves testing of the sensor with the help of an Arduino program. Fig. 5 shows the screenshot of the code used for detecting liquid.

An integer 'x' is defined which stores the value of the input read from the output of the sensor. Whenever the value of the integer becomes greater than zero, it indicates detection of the liquid. Therefore, an if/else condition is used to trigger buzzer output.

When value of integer x = 0, the serial monitor prints NO LIQUID and output of the buzzer is kept low. When integer x is not equal to 0, the output of buzzer is high and LIQUID DETECTED is printed on serial monitor.

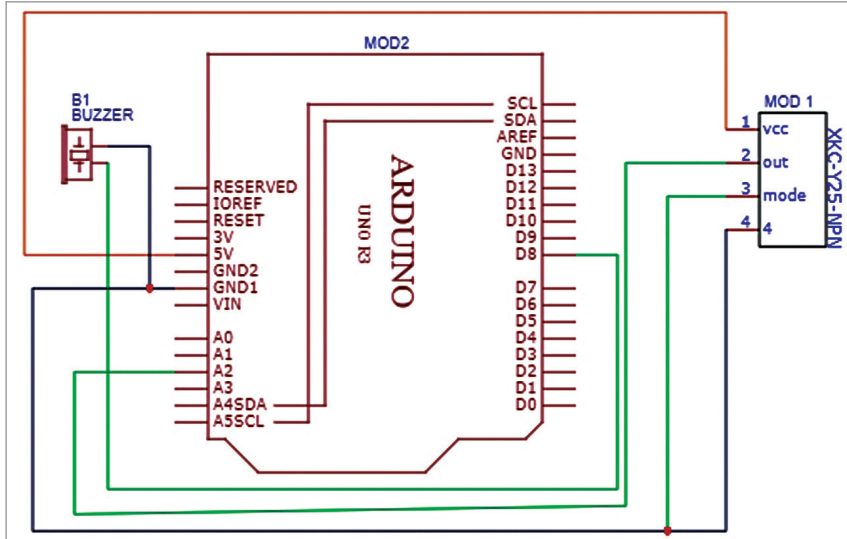


Fig. 3: Circuit diagram

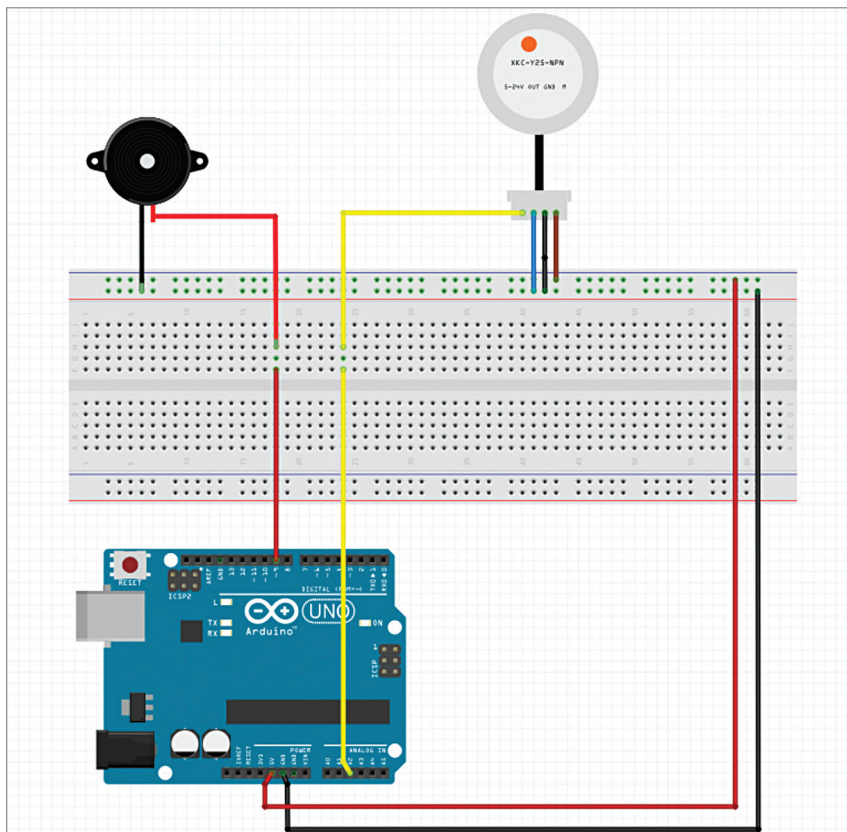


Fig. 4: Wiring of the detector on breadboard

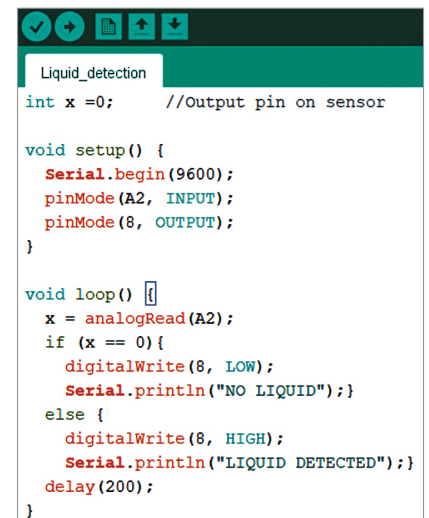


Fig. 5: Screenshot of the source code